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MTH422A
Assignment 2

classmate
Date
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1) $X_1, \dots, X_n \mid \mu, \sigma^2 \overset{iid}{\sim} N(\mu, \sigma^2)$

$\sigma^2 \sim IG(a, b)$

$\mu \mid \sigma^2 \sim N(0, c\sigma^2)$

The posterior distribution $\pi(\mu, \sigma^2 \mid X_1, \dots, X_n)$ is given by

$$\pi(\mu, \sigma^2 \mid X_1, \dots, X_n) \propto \pi(X_1, \dots, X_n \mid \mu, \sigma^2)\, \pi(\mu, \sigma^2)$$

$$= \pi(X_1, \dots, X_n \mid \mu, \sigma^2)\, \pi(\mu \mid \sigma^2)\, \pi(\sigma^2)$$

$$\propto (\sigma^2)^{-n/2} \exp\left[-\frac{1}{2}\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{\sigma^2}\right](\sigma^2)^{-1/2}\exp\left[-\frac{\mu^2}{2c\sigma^2}\right]$$

$$\times (\sigma^2)^{-a-1}\exp\left[\frac{-b}{\sigma^2}\right]$$

$$= (\sigma^2)^{-a-\frac{n}{2}-\frac{3}{2}} \exp\left[\frac{-1}{\sigma^2}\left(\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)\right]$$

$$= (\sigma^2)^{-\underbrace{\left(\frac{2a+n+1}{2}\right)}_{A}-1} \exp\left[-\frac{1}{\sigma^2}\underbrace{\left(\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)}_{B}\right]$$

This is the kernel of an Inverse Gamma distribution with shape $A = a + \frac{n}{2} + \frac{1}{2}$

& rate $B = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b$

To find the marginal posterior $\pi(\mu \mid x_1, \ldots, x_n)$ we integrate over σ^2

$$\pi(\mu \mid x_1, \ldots, x_n) \propto \int_0^{\infty} (\sigma^2)^{-(a+\frac{n}{2}+\frac{1}{2})-1} \exp\left[-\frac{B}{\sigma^2}\right] d\sigma^2$$

$$= \int_0^{\infty} (\sigma^2)^{-A-1} \exp\left[-\frac{B}{\sigma^2}\right] d\sigma^2$$

$$\propto B^{-A}$$

$$= \left(\sum_{i=1}^{n} \frac{(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)^{-(a+\frac{n}{2}+\frac{1}{2})}$$

$$= \left(\sum_{i=1}^{n} \frac{(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)^{-\frac{(2a+n+1)}{2}}$$

It is similar to the kernel of Student's t distribution with degrees of freedom $2a+n+1$

The posterior predictive distribution of x_{n+1}, i.e $\pi(x_{n+1} \mid x_1, \ldots, x_n)$ is given by

$$PPD = \pi(x_{n+1} \mid x_1, \ldots, x_n)$$

$$= \int_0^{\infty}\int_{-\infty}^{\infty} \pi(x_{n+1} \mid \mu, \sigma^2)\, \pi(\mu, \sigma^2 \mid x_1, \ldots, x_n)\, d\mu\, d\sigma^2$$

$$\propto \int_{-\infty}^{\infty}\int_0^{\infty} (\sigma^2)^{-1/2} \exp\left[-\frac{(x_{n+1}-\mu)^2}{2\sigma^2}\right] (\sigma^2)^{-(a+\frac{n+1}{2})-1}$$
$$\times \exp\left[-\frac{1}{\sigma^2}\left(\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)\right] d\sigma^2\, d\mu$$

$$= \int_{-\infty}^{\infty}\int_{0}^{\infty} (\sigma^{2})^{-(a+\frac{n}{2}+1)-1} \exp\left[-\frac{1}{\sigma^2}\left(\frac{\sum_{i=1}^{n+1}(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)\right] d\sigma^2 \, d\mu$$

$$\propto \int_{-\infty}^{\infty} \left(\frac{\sum_{i=1}^{n+1}(x_i-\mu)^2}{2} + \frac{\mu^2}{2c} + b\right)^{-\frac{(2a+n+2)}{2}} d\mu$$

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2) $X_1, \ldots, X_m \mid \mu_1, \sigma^2 \overset{IID}{\sim} N(\mu_1, \sigma^2)$

$Y_1, \ldots, Y_n \mid \mu_2, \sigma^2 \overset{IID}{\sim} N(\mu_2, \sigma^2)$

$\sigma^2 \sim IG(a, b)$.

$\mu_1, \mu_2 \mid \sigma^2 \overset{IID}{\sim} N(0, c\sigma^2)$

So, $\mu_1 - \mu_2 \mid \sigma^2 \sim N(0, 2c\sigma^2)$

The posterior distribution $\pi(\mu_1 - \mu_2, \sigma^2 / X_1, \ldots, X_m, Y_1, \ldots, Y_n)$ is given by

$$\pi(\mu_1-\mu_2, \sigma^2 / X_1, \ldots, X_m, Y_1, \ldots, Y_n) = \pi(X_1, \ldots, X_m \mid \mu_1, \sigma^2) \times \pi(Y_1, \ldots, Y_n \mid \mu_2, \sigma^2)\, \pi(\mu_1 - \mu_2, \sigma^2)$$

$$\propto (\sigma^2)^{-m/2} \exp\left[-\frac{\sum_{i=1}^{m}(x_i-\mu_1)^2}{2\sigma^2}\right] (\sigma^2)^{-n/2} \exp\left[-\frac{\sum_{j=1}^{n}(y_j-\mu_2)^2}{2\sigma^2}\right]$$
$$\times (\sigma^2)^{-1/2} \exp\left[-\frac{(\mu_1-\mu_2)^2}{4c\sigma^2}\right] \times (\sigma^2)^{-a-1} \exp\left[-\frac{b}{\sigma^2}\right]$$

$$= (\sigma^2)^{\underbrace{-\left(\frac{m}{2}+\frac{n}{2}+\frac{1}{2}+a\right)}_{A}-1} \exp\left[-\frac{1}{\sigma^2}\underbrace{\left(\sum_{i=1}^{m}\frac{(x_j-\mu_1)^2}{2} + \sum_{j=1}^{n}\frac{(y_j-\mu_2)^2}{2} + \frac{(\mu_1-\mu_2)^2}{4c} + b\right)}_{B}\right]$$

This has same kernel as an Inverse Gamma distribution with shape $A = a + \frac{m+n+1}{2}$

$$\& B = \sum_{i=1}^{m}\frac{(x_i-\mu_1)^2}{2} + \sum_{j=1}^{n}\frac{(y_j-\mu_2)^2}{2} + \frac{(\mu_1-\mu_2)^2}{4c} + b$$

To find $\pi(\mu_1 - \mu_2 \mid x_1, \ldots, x_m, y_1, \ldots, y_m)$
we integrate over σ^2

$$\pi(\mu_1 - \mu_2 \mid x_1, \ldots, x_m, y_1, \ldots, y_m) = \int_0^\infty (\sigma^2)^{-A-1} \exp\left(-\frac{B}{\sigma^2}\right) d\sigma^2$$

$$\propto B^{-A}$$

$$= \left(\sum_{i=1}^{m} \frac{(x_i - \mu_1)^2}{2} + \sum_{j=1}^{m} \frac{(y_i - \mu_2)^2}{2} + \frac{(\mu_1 - \mu_2)^2}{4c} + b\right)^{-\left(\frac{2a+m+n+1}{2}\right)}$$

Assuming putting $m = n$

$$= \left(\sum_{i=1}^{m} \frac{(x_i - \mu_1)^2}{2} + \sum_{i=1}^{m} \frac{(y_i - \mu_2)^2}{2} + \frac{(\mu_1 - \mu_2)^2}{4c} + b\right)^{-\left(\frac{2a+2m+1}{2}\right)}$$

$$= \left(\sum_{i=1}^{m} \frac{(x_i - \mu_1)^2 + (y_i - \mu_2)^2}{2} + \frac{(\mu_1 - \mu_2)^2}{4c} + b\right)^{-\left(\frac{2a+2m+1}{2}\right)}$$

It is similar to the kernel of student's t distribution with degrees of freedom $2a + 2m + 1$.

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3) $X_1, \dots, X_m \overset{IID}{\underset{\lambda_1}{\sim}} \text{Poisson}(\lambda_1)$

$Y_1, \dots, Y_m \overset{IID}{\underset{\lambda_2}{\sim}} \text{Poisson}(\lambda_2)$

$\lambda_1, \lambda_2 \overset{IID}{\sim} \text{Gamma}(a, b)$

$$\pi(\lambda_1, \lambda_2 \mid X_1, \dots, X_m, Y_1, \dots, Y_m)$$

$$\propto \lambda_1^{\sum_{i=1}^m x_i} \exp\left(-m \sum_{i=1}^m x_i\right) \lambda_2^{\sum_{j=1}^m y_j} \exp\left(-m \sum_{j=1}^m y_j\right) \lambda_1^{a-1} \exp(-b\lambda_1) \; \lambda_2^{a-1} \exp(-b\lambda_2)$$

$$= \lambda_1^{a+\sum_{i=1}^m x_i - 1} \exp\left(-(m+b)\sum_{i=1}^m x_i\right) \lambda_2^{b+\sum_{j=1}^m y_j - 1} \exp(-(b+m)\lambda_2$$

$$= \pi(\lambda_1 \mid X_1, \dots, X_m) \; \pi(\lambda_2 \mid Y_1, \dots Y_m)$$

$\lambda_1 \mid X_1 \dots X_m \sim \text{Gamma}\left(a + \sum_{i=1}^m X_i, b+m\right)$

$\lambda_2 \mid Y_1, \dots, Y_m \sim \text{Gamma}\left(a + \sum_{i=1}^m X_i, b+m\right)$

Assuming $m = n$

$$\theta = \lambda_1/(\lambda_1 + \lambda_2) \sim \text{Beta}\left(a + \sum_{i=1}^n X_i, a + \sum_{i=1}^n Y_i\right)$$

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4) $X_1 \sim N(0,1)$
$x_{t+1} \mid x_t \sim N(\rho x_t, 1-\rho^2)$
$t = 1, 2, \dots, T$

$\rho \sim \text{Uniform}(-1, 1)$

$$\pi(X_1, \dots, X_T \mid \rho) = \pi(x_T \mid X_1, \dots x_{T-1}, \rho) \times \pi(X_1, \dots x_{T-1} \mid \rho)$$

$\vdots$

$$= \pi(x_T \mid x_1, \dots, x_{T-1}, \rho)\, \pi(x_{T-1} \mid x_1, \dots x_{T-2}, \rho) \cdots \pi(x_1 \mid \rho)$$

$$= \prod_{t=2}^{T} \pi(x_t \mid x_1, \dots, x_{t-1}; \rho)\ \pi(x_1)$$

$$= \prod_{t=2}^{T} \pi(x_t \mid x_{t-1}, \rho)\ \pi(x_1)$$

$$\propto (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left[-\sum_{t=2}^{T} \frac{(x_t - \rho x_{t-1})^2}{2(1-\rho^2)}\right] \exp\left[-\frac{x_1^2}{2}\right]$$

$$= (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left[\frac{-1}{2}\left(\sum_{t=2}^{T} \frac{(x_t - \rho x_{t-1})^2}{1-\rho^2} + x_1^2\right)\right]$$

The posterior distribution $\pi(\rho \mid X_1, \dots, X_T)$ is given by

$$\pi(\rho \mid X_1 \dots X_T) \propto (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left[-\frac{1}{2}\left(\sum_{t=2}^{T} \frac{(x_t - \rho x_{t-1})^2}{1-\rho^2}\right)\right] \times I(\rho \in [-1, 1])$$

$$= (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left[-\frac{1}{2}\left(\sum_{t=2}^{T} \frac{(x_t - \rho x_{t-1})^2}{1-\rho^2}\right)\right] I(\rho \in [-1, 1])$$

the posterior predictive density is given by

$$\pi(x_{T+1}, x_{T+2} \mid x_1, \ldots, x_T) = \int_{-1}^{1} \pi(x_{T+1}, x_{T+2} \mid \rho)\, \pi(\rho \mid x_1, \ldots x_T)\, d\rho$$

$$\propto \int_{-1}^{1} (1-\rho^2)^{-2/2} \exp\left[-\frac{(x_{t+2}-\rho x_{t+1})^2}{2(1-\rho^2)} - \frac{(x_{t+1}-\rho x_t)^2}{2(1-\rho^2)} \right]$$

$$\times (1-\rho^2)^{-\left(\frac{T-1}{2}\right)} \exp\left[-\sum_{t=2}^{T} \frac{(x_t - \rho x_{t-1})^2}{2(1-\rho^2)} \right] d\rho$$

$$= \int_{-1}^{1} (1-\rho^2)^{\frac{-(T+1)}{2}} \exp\left[\frac{-1}{2} \left(\sum_{t=2}^{T+2} \frac{(x_t - \rho x_{t-1})^2}{2(1-\rho^2)} \right) \right] d\rho$$